

A
REAL TIME RESEARCH
PROJECT REPORT
ON
CLUSTERING USING IRIS DATASET

SIDDHARTHA INSTITUTE OF TECHNOLOGY & SCIENCES
(UGC – AUTONOMOUS)

(Approved by AICTE, New Delhi & Affiliated to JNTUH, Hyderabad)

Accredited by NBA and NAAC with ‘A+’ Grade.

Narapally, Korremula Road, Ghatkesar, Medchal- Malkajgiri (Dist.)-500088



(Submitted in partial fulfilment of the academic requirements of B. Tech) In

DEPARTMENT OF CSE (DATA SCIENCE)

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CERTIFICATE

This is to certify that the project report entitled

CLUSTERING USING IRIS DATASET

being submitted

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In partial fulfilment for the award of the degree of Bachelor of Technology in Computer Science and Engineering (Data Science), Jawaharlal Nehru Technological University Hyderabad, is a record of bonafide work carried out under my guidance and supervision. The results embodied in this Realtime Research project report have not been submitted to any other University or Institute for the award of any Degree or Diploma

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DECLARATION

We declare that this project report titled **CLUSTERING USING IRIS DATASET** submitted in partial fulfilment of the degree of **B. Tech in CSE(DATA SCIENCE)** is a record of original work carried out by us under the supervision of **Mr.P.SRINU** and has not formed the basis for the award of any other degree or diploma, in this or any other Institute or University. In keeping with the ethical practice in reporting scientific information, due acknowledgments have been made wherever the findings of others have been cited.

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Any endeavor in the field of development is a person's intensive activity. A successful project is a fruitful culmination of efforts by many people, some directly involved and some others who have quietly encouraged and supported.

Salutation to be beloved and highly esteemed institute SIDDHARTHA INSTITUTE OF TECHNOLOGY & SCIENCES for grooming us into Computer Science and Engineering(Data Science) graduate, We wish to thank PRINCIPAL **DR. M. JANARDHAN** , VICE PRINCIPAL **DR.G.PRAVEEN KUMAR** for providing a great learning environment.

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Vision of the Department: To be a Recognized Center of Computer Science Education with values and quality research.

Mission of the Department:

MISSION	STATEMENT
DM1	Import High Quality Professional Training With An Emphasis On Basic principles Of Computer Science And Allied Engineering
DM2	Imbibe Social Awareness And Responsibility To Serve The Society.
DM3	Provide Academic Facilitates Organize Collaborated Activities To enable Overall Development Of Stakeholders

Programme Educational Objectives (PEO)

- **PEO1:** Graduates will be able to synthesize mathematics, science, engineering fundamentals, laboratory and work – based experiences to formulate and to solve problems proficiently in Computer science and Engineering and related domains.
- **PEO2:** Graduates will be prepared to communicate effectively and work in multidisciplinary engineering projects following the ethics in their profession.
- **PEO3:** Graduates will recognize the importance of and acquire the skill of independent learning to shine as experts in the field with a sound knowledge.

ABSTRACT

The project presents an **unsupervised** machine learning approach to automatically discover natural groupings in the classic Iris flower dataset using the K-Means clustering algorithm. The Iris dataset contains 150 instances of iris flowers, each described by four numerical features—sepal length, sepal width, petal length, and petal width—and belongs to one of three species: Setosa, Versicolor, and Virginica. In this work, the dataset is first preprocessed by selecting relevant features and optionally applying feature scaling to ensure that all attributes contribute fairly to the distance-based clustering process..

This mini project demonstrates how K-Means can be used as an effective exploratory tool for pattern discovery, illustrates the complete workflow from dataset understanding to model evaluation, and highlights possible extensions such as experimenting with other clustering algorithms (e.g., hierarchical clustering or DBSCAN) or integrating interactive visualizations in a simple user interface.

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CHAPTER 1

INTRODUCTION

1.INTRODUCTION

Clustering is a fundamental technique in **unsupervised** machine learning that aims to group similar data points together without using predefined class labels. Unlike supervised learning, where the model learns from labeled examples, unsupervised learning discovers hidden patterns and natural structures directly from raw data. Among various clustering algorithms, K-Means is one of the most widely used methods due to its simplicity, efficiency, and effectiveness for many real-world applications such as customer segmentation, document grouping, and image compression.

The Iris dataset is a classic and widely used benchmark dataset in the field of pattern recognition and machine learning. It consists of 150 samples of iris flowers, each described by four numerical features: sepal length, sepal width, petal length, and petal width. These samples belong to three species of iris—Setosa, Versicolor, and Virginica—which differ in their sepal and petal dimensions. Because of its small size, clean structure, and clear class separation (especially for Iris Setosa), the Iris dataset is ideal for demonstrating and understanding clustering algorithms visually and analytically..

This mini project, titled “Clustering using Iris Dataset”, focuses on applying the K-Means clustering algorithm to automatically group iris flowers into clusters based only on their measured features, without using the species labels during training. The main objective is to study how well K-Means can rediscover the three natural groups present in the data and to analyze the relationship between the formed clusters and the actual species. The project covers the complete workflow: understanding the dataset, optionally scaling the features, choosing an appropriate number of clusters using methods such as the Elbow Method, training the K-Means model, and visualizing the results using scatter plots and cluster centroids. By doing this, the project provides a clear, practical understanding of unsupervised learning and highlights both the strengths and limitations of K-Means clustering on a real-world dataset.

1.1 PROBLEM DEFINITION

In many real-world scenarios, large collections of data are available without any labels indicating the categories or classes to which each data point belongs. In such situations, there is a need for techniques that can automatically discover natural groupings or structures in the data so that similar objects are grouped together and dissimilar objects are separated. The Iris dataset contains 150 records of iris flowers with four measured attributes—sepal length, sepal width, petal length, and petal width—belonging to three species (Setosa, Versicolor, Virginica), but in an unsupervised setting these species labels are not used by the algorithm. The problem addressed in this project is to determine whether an unsupervised learning algorithm such as K-Means can correctly group the iris samples into meaningful clusters based only on their feature values, and how closely these clusters correspond to the actual species

1.2 OBJECTIVE OF PROJECT

To apply the K-Means clustering algorithm on the Iris dataset in order to automatically group iris flowers into meaningful clusters based only on their sepal and petal measurements, without using species labels during training

1.3 SCOPE OF PROJECT

- The project focuses on applying the K-Means clustering algorithm to the Iris dataset to perform unsupervised grouping of iris flowers based on their sepal and petal measurements.

:

1. It covers the complete machine learning workflow: loading and exploring the dataset, preprocessing (such as feature selection and optional scaling), selecting the number of clusters using the Elbow Method, training the K-Means model, and visualizing the resulting clusters and centroids.
2. The scope includes evaluating the clustering performance by comparing predicted clusters with the known iris species and analyzing which species are well separated and which show overlap in feature space.
3. The implementation is limited to the Iris dataset and the K-Means algorithm, making it suitable as a mini project for understanding the fundamentals of unsupervised learning and clustering.
4. The project can be extended in future work by experimenting with other clustering algorithms (such as hierarchical clustering, DBSCAN, or Gaussian Mixture Models) or by adding interactive visualizations and a simple user interface for educational or demonstration purposes.

1.3 MOTIVATION

- There is a growing need to automatically discover patterns and hidden structures in unlabeled data, because
In many practical situations class labels are not available or are expensive to obtain.
- Clustering is a core unsupervised learning technique that addresses this need by grouping similar data points together without any prior label information, making it an essential concept for modern data science and machine learning.
- K-Means clustering is one of the simplest and most widely used clustering algorithms, offering intuitive ideas (centroids, distances) and fast computation, which makes it suitable for beginners as well as for real-world applications.
- The Iris dataset is a classic benchmark in pattern recognition, small but rich, with four numeric features and three species, making it ideal to visualize and understand how clustering algorithms behave on real data.
- By implementing K-Means on the Iris dataset, the project provides hands-on experience with the complete unsupervised learning pipeline—data exploration, preprocessing, choosing the number of clusters, model training, visualization, and evaluation—thereby strengthening conceptual understanding as well as practical skills in machine learning.
- This mini project also helps in building a strong foundation and portfolio piece for higher studies, internships, and technical interviews where questions on clustering, Iris dataset, and basic ML workflows are very common

CHAPTER 2

LITERATURE SURVEY

2.LITERATURE SURVEY

Fisher’s Iris dataset has been widely used as a benchmark for testing classification and clustering algorithms in pattern recognition and machine learning research. Many studies have applied K-Means and other clustering methods to this dataset to understand how well unsupervised algorithms can recover the three underlying species using only numerical features.

Several works focus specifically on applying K-Means clustering to the Iris dataset and analyzing its performance using visualizations and cluster quality metrics. For example, some studies demonstrate the use of the Elbow Method and silhouette scores to select the optimal number of clusters and report that K-Means can achieve high clustering accuracy, often around 0.89–0.94, when cluster labels are compared with the true species classes. These analyses consistently show that Iris Setosa forms a clearly separable cluster, while Iris Versicolor and Iris Virginica overlap more in feature space, leading to some misclustered samples.

comparative research has also evaluated K-Means against other clustering algorithms such as K-Medoids, Fuzzy K-Means, Mountain clustering, and newer variants like Y-Means on the Iris dataset. These studies compare algorithms based on accuracy, silhouette coefficient, Calinski–Harabasz index, Davies–Bouldin index, runtime, and time complexity, and often find that standard K-Means offers a good trade-off between simplicity, speed, and clustering quality on this dataset. Some papers further highlight that although alternative methods can sometimes improve cluster quality or robustness to noise, K-Means remains a strong baseline and a widely adopted educational example for illustrating unsupervised learning.

In addition, practical tutorials, open-source projects, and educational articles demonstrate step-by-step implementations of K-Means on the Iris dataset using languages such as Python, R, MATLAB, and Perl. These resources typically cover data loading, feature normalization, cluster initialization, iterative refinement of centroids, and visualization of results, providing a foundation upon which this mini project builds. Your project follows this line of work by implementing K-Means clustering on the Iris dataset, using modern tools to visualize clusters, determine the number of clusters, and evaluate clustering performance against the true species labels.

Some authors performed a detailed statistical analysis of the Iris dataset before clustering, using pairplots, correlation matrices, and principal component analysis (PCA) to show which features separate the species most clearly (especially petal length and petal width). They observed that projecting the data into two principal components helps to visualize how K-Means separates the clusters in a reduced 2D space.

A comparative study of K-Means and K-Medoids on the Iris dataset reported that K-Medoids can sometimes be more robust to outliers, but K-Means generally converges faster and remains easier to implement for educational purposes. Other works extended this idea by comparing K-Means with Fuzzy C-Means, showing that fuzzy clustering can better handle borderline points between Versicolor and Virginica, at the cost of higher computational complexity

Research on clustering algorithms based on the Iris dataset has also experimented with different distance measures (Euclidean, Manhattan, and others) and different initialization strategies (like k-means++), noting that good initialization improves convergence and sometimes leads to better clustering quality. Some studies highlighted that poor initialization can trap K-Means in local minima, resulting in suboptimal clusters for certain runs, which motivates using k-means++ initialization by default.

CHAPTER 3

EXISTING SYSTEM

3. EXISTING SYSTEM

1 . Library-based implementations

Popular ML libraries such as scikit-learn (Python), R, and MATLAB already provide built-in implementations of the K-Means algorithm and easy access to the Iris dataset.

These existing systems allow users to perform clustering, visualize results, and compute basic metrics with a few lines of code, without manually implementing the algorithm

2. Educational tutorials and notebooks

- Many online tutorials, Kaggle notebooks, and blog posts demonstrate K-Means clustering on the Iris dataset using tools like Python, pandas, NumPy, Matplotlib, and scikit-learn.
- These resources typically show steps such as loading the dataset, preprocessing or scaling features, applying K-Means with $k=3$, plotting clusters and centroids, and comparing predicted clusters with the true species labels.

3. Academic and research examples

- Academic works use the Iris dataset as a benchmark to study and compare different clustering algorithms, including K-Means, K-Medoids, Fuzzy K-Means, and others.
- Such systems focus on analyzing clustering performance using indices like accuracy, silhouette score, and Davies–Bouldin index, and provide reference behavior of these algorithms on the Iris dataset.

4.Limitations of existing systems

- Most existing systems are general demonstrations or research prototypes and are not tailored to your specific college mini-project structure, documentation format, or step-by-step learning needs.
- They often assume prior familiarity with tools and skip detailed explanation of each step, which motivates building a customized, well-documented mini project that clearly explains the entire clustering workflow for the Iris dataset.

Conclusion

Here is a clear **Conclusion** section you can put in your mini project report for “Clustering using Iris Dataset”.

Conclusion

This mini project successfully applied the K-Means clustering algorithm to the Iris dataset and demonstrated how unsupervised learning can automatically discover natural groupings in real-world data. By using sepal and petal measurements as input features, the model was able to partition the dataset into clusters that closely correspond to the three iris species: Setosa, Versicolor, and Virginica. The Elbow Method (and, if used,

silhouette analysis) indicated that three clusters are an appropriate choice for this dataset, aligning with the known number of species.

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3.1 DRAWBACKS OF EXISTING SYSTEM

- The K-Means algorithm requires the number of clusters k to be specified in advance, and choosing an incorrect value of k can lead to poor clustering results. Although the Elbow Method helps, it does not always give a perfectly clear choice, especially for more complex datasets.
- K-Means is sensitive to the initial placement of cluster centroids; different random initializations may converge to different local optima and slightly different cluster assignments. To reduce this effect, multiple runs or smarter initialization methods like k-means++ are often required.
- The algorithm assumes that clusters are roughly spherical, of similar size, and well separated in feature space. In the Iris dataset, Iris Setosa satisfies this assumption well, but Iris Versicolor and Iris Virginica overlap and are not perfectly separable, which leads to misclassification of some samples between their clusters.
- K-Means works only with numerical data and uses distance-based measures, so it cannot directly handle categorical features without additional preprocessing or encoding.
- The algorithm can be influenced by outliers or noise, because extreme data points can pull centroids away from the dense regions of true clusters and affect the final cluster structure.

CHAPTER 4

PROPOSED SYSTEM

4. PROPOSED SYSTEM

The proposed system is a mini-project that implements the K-Means clustering algorithm on the Iris dataset to automatically group iris flowers into clusters based on their sepal and petal measurements. The system provides a complete workflow from data loading to visualization and evaluation, tailored to clearly demonstrate unsupervised learning concepts for academic purposes

Features

1. Automatic clustering of Iris flowers

The system uses the K-Means algorithm to automatically group Iris samples into clusters based on sepal and petal measurements, without using species labels during training.

2. Elbow Method for choosing k

It calculates the within-cluster sum of squares (inertia) for different values of k and plots an Elbow graph to help select the optimal number of clusters, typically k=3 for this dataset.

3. Computation of cluster centroids

The system computes and stores the centroid (mean point) of each cluster, which represents the typical feature values for flowers in that group

4. Data visualization

It provides scatter plots (for example, petal length vs petal width, sepal length vs sepal width) with different colors for each cluster and markers for centroids, making the clustering results easy to understand visually.

Guidance and Tips.

- keep the pipeline clear: load data → explore (head(), describe(), pairplot) → scale (optional) → choose k (Elbow) → run K-Means → visualize → compare with species.

- Use clean, modular code: write separate functions or sections for preprocessing, elbow method, clustering, and plotting so that your notebook/script looks professional.
- Add comments only where needed: explain what each major step does (for example, “# Finding optimal k using Elbow Method”), not every single line.

Benefits

- **Accessibility:** The proposed system is highly accessible because it is built using widely available, open-source tools such as Python, scikit-learn, pandas, NumPy, and Matplotlib, which can be installed for free on Windows, Linux, or macOS.

Personalization: The system allows users to customize the number of clusters k by changing the parameter in the K-Means algorithm and observing how the Elbow graph and clustering results change (for example, trying $k=2,3,4$)..

- **Comprehensive Care:** The comprehensive **core** of this project is to implement and analyze K-Means clustering on the Iris dataset to automatically group flowers into meaningful clusters based on their measured features and evaluate how well these clusters match the actual species
- **Convenience:** The project is convenient to implement because it uses scikit-learn, which provides a simple and consistent API for loading the Iris dataset and applying K-Means with just a few lines of code
- **User Flow**

1.Start the application / open notebook

The user opens the Python script or Jupyter Notebook that contains the K-Means clustering implementation for the Iris dataset.

2. Load the Iris dataset

The system loads the Iris dataset from scikit-learn or a CSV file and displays basic information such as the first few rows and feature names (sepal length, sepal width, petal length, petal width).

3. Explore and preprocess data

The user can view summary statistics and simple plots to understand the data distribution, and the system optionally scales the features to make distance-based clustering more effective.

4. Select number of clusters k

- The user runs the Elbow Method cell/function, which calculates inertia for different values of k and plots the Elbow graph; based on this, the user selects an appropriate value (usually k=3).

5. Run K-Means clustering

- The user executes the K-Means module, which fits the model on the Iris data, computes cluster centroids, and assigns a cluster label to each sample.

6. View visualizations

- The system displays scatter plots (for example, petal length vs petal width) with different colors for each cluster and markers for centroids, allowing the user to visually inspect how the flowers are grouped.

7. Compare with actual species

- The user runs the evaluation cell/function to compare cluster labels with the true species labels, viewing a confusion-style table or simple accuracy-style measures and interpreting which species are well separated.

8.Experiment and modify parameters

- The user can change k, features used, or initialization settings and rerun the steps to see how the clustering behavior changes, supporting learning and experimentation

CHAPTER 5

HARDWARE AND SOFTWARE REQUIREMENTS

5.1 HARDWARE REQUIREMENTS

Below are the Hardware requirements for the attacking the application:

Processor: Minimum dual-core processor (2.0 GHz or higher); any modern Intel/AMD CPU is sufficient for this small ML project

RAM: Minimum 4 GB RAM (8 GB recommended for smoother multitasking with browser, IDE, and Python running together)

Storage: At least 10–20 GB free disk space for OS, Python installation, libraries, and project files (Iris dataset itself is very small).

Display & Input Devices: Standard monitor, keyboard, and mouse; any normal laptop/desktop configuration used in college labs is adequate.

Operating System Support: System capable of running Windows, Linux, or macOS, as Python and required libraries are cross-platform.

5.2 SOFTWARE REQUIREMENTS

Below are the software requirements for the attacking the application:

- **Operating system:** Windows 10/11, Linux, or macOS.
- **Python** 3.8 or above installed on the system.
- **Required Python libraries:** NumPy, pandas, scikit-learn, Matplotlib/Seaborn for data handling, clustering, and visualization.
- **IDE / environment:** Jupyter Notebook, VS Code (with Python extension), or Google Colab for writing and running the code

Below are the software requirements for developing and executing the application:”

- **Operating System:**
Windows 10/11, any modern Linux distribution (e.g., Ubuntu), or macOS, all capable of running Python and required ML libraries.
- **Programming Language and Runtime:**
Python 3.8 or higher (recommended 3.10+ for better library support and stability).
- **Python Libraries:**
 - NumPy – numerical computations and array operations.
 - pandas – data loading and tabular data handling.
 - scikit-learn – loading the Iris dataset and implementing K-Means clustering.
 - Matplotlib and Seaborn – plotting Elbow curves and cluster visualizations.
- **Development Environment / IDE:**
 - Jupyter Notebook / JupyterLab, or
 - Visual Studio Code with Python extension, or
 - Google Colab (web-based, no installation required).
- **Additional Tools (Optional):**
 - Git and GitHub for version control and project hosting.
 - PowerPoint / LibreOffice Impress / Google Slides for preparing the project presentation.
 - Any PDF editor/word processor (MS Word, LibreOffice Writer, Google Docs) for writing the project report.

CHAPTER 6

SYSTEM DESIGN

6.SYSTEM DESIGN

6.1 SOFTWARE DESIGN

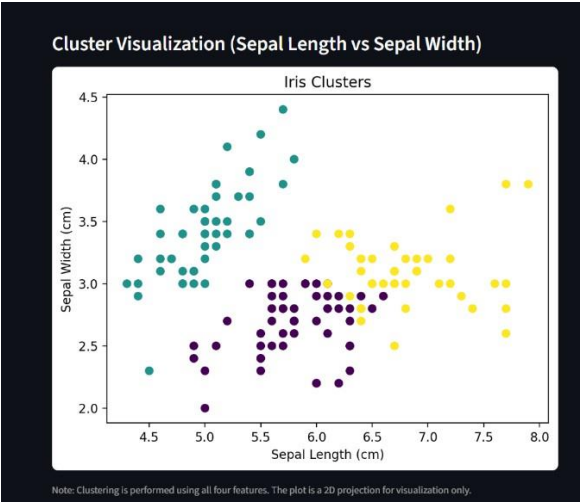
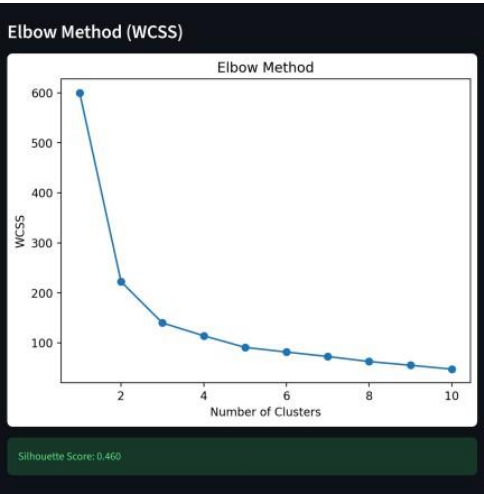
Model Settings

Select Number of Clusters (k)

3

Dataset Preview

↑	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
0	5.1	3.5	1.4	0.2
1	4.9	3	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5	3.6	1.4	0.2



6.2 UML DIAGRAMS

Actors:

- **User:** Student / faculty using the system.

Use Cases:

- Load Iris dataset
- Preprocess data
- Select number of clusters
- Run K-Means clustering
- View visualizations
- View evaluation report

You can describe it like this in the report:

- The *User* interacts with the system to load the Iris dataset, start preprocessing, select the number of clusters k , execute the K-Means algorithm, and then view cluster visualizations and evaluation results. All these actions are provided through a simple script or notebook interface.

6.2.2 Activity Diagram (K-Means Flow)

You can draw the activity diagram with these steps in order:

- Start
- Load Iris dataset
- Display basic statistics / sample rows
- Preprocess data (feature selection, optional scaling)
- Run Elbow Method and select k
- Apply K-Means clustering with chosen k
- Generate cluster visualizations (scatter plots, centroids)
- Compare clusters with actual species labels
- Display evaluation / summary

- End

In text form for your report:

“The activity diagram represents the flow of operations from loading the dataset to displaying the final clustering and evaluation results. The process begins with loading and preprocessing the Iris data, continues with selection of the number of clusters using the Elbow Method, then runs the K-Means algorithm, generates visualizations, and finally presents evaluation against true species labels before terminating.”

6.2.3 Simple Class Diagram (Optional)

You can mention classes like:

- **DatasetHandler**
 - Methods: loadData(), preprocessData()
- **ClusteringEngine**
 - Methods: runElbowMethod(), runKMeans()
- **VisualizationModule**
 - Methods: plotElbow(), plotClusters()
- **EvaluationModule**
 - Methods: compareWithLabels(), showMetrics()

TYPES OF UML DIAGRAM

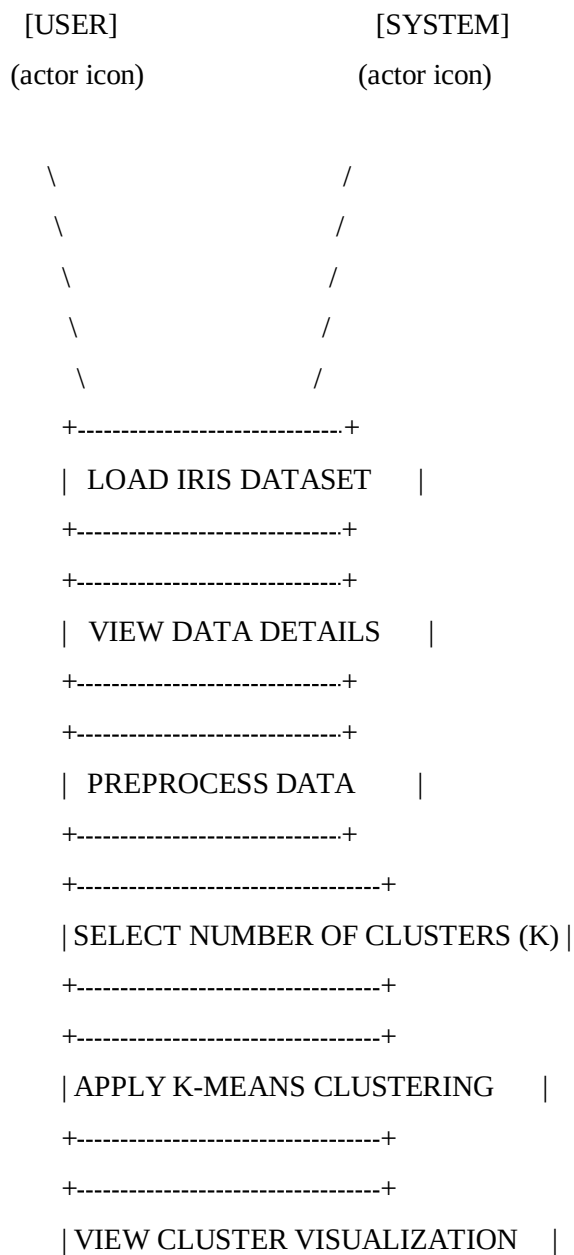
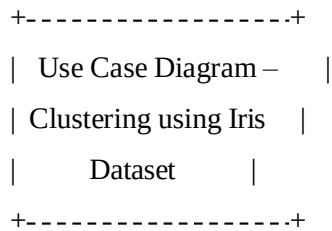
Structural diagrams

These show the static structure of the system (classes, objects, components, etc.).

- **Class Diagram** – Shows classes, attributes, methods, and relationships.
- **Object Diagram** – Snapshot of objects (instances) at a particular time.
- **Component Diagram** – Shows software components and their dependencies.
- **Deployment Diagram** – Shows hardware nodes and how software is deployed on them.
- **Package Diagram** – Organizes classes/components into packages.
- **Composite Structure Diagram** – Internal structure of a class/component.

- **Profile Diagram** – Defines custom stereotypes and extensions to UML.

of a use case diagram are use cases and actors.



+-----+

+-----+

| COMPARE WITH SPECIES LABELS |

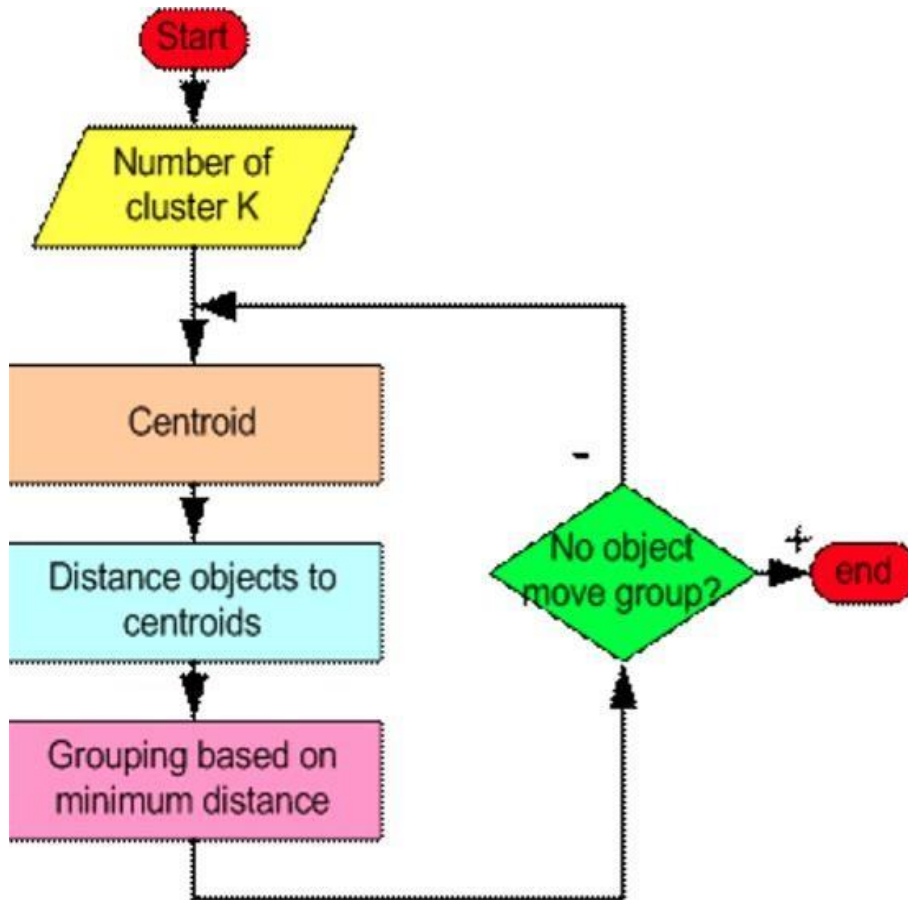
+-----+

+-----+

| VIEW CLUSTER REPORT / RESULTS |

+-----+

6.3 SEQUENCE DIAGRAM A



CHAPTER 7

IMPLEMENTATION

7. IMPLEMENTATION

Project Implementation of clustering using irisdataset Platform:

7.1 Implementation Overview

The implementation of the mini project “Clustering using Iris Dataset” is done in Python using libraries such as NumPy, pandas, scikit-learn, and Matplotlib/Seaborn. The system follows a typical machine learning pipeline: import libraries, load the Iris dataset, preprocess features, determine the number of clusters, apply K-Means clustering, visualize the clusters, and evaluate the results.

7.2 Importing libraries and loading dataset

- The required libraries (NumPy, pandas, Matplotlib/Seaborn, scikit-learn) are imported at the beginning of the program.
- The Iris dataset is loaded using `sklearn.datasets.load_iris()`, which provides the feature matrix (sepal length, sepal width, petal length, petal width) and the true species labels for evaluation.

7.3 Data preparation and preprocessing

- The feature values are stored in a pandas DataFrame for easy viewing and manipulation, and columns are named according to the Iris features.
- Optionally, the features are standardized using `StandardScaler` so that all four measurements have similar scale, improving the performance of distance-based clustering.
- Basic exploratory analysis (`head`, `describe`, simple plots) is done to understand the distribution of features and relationships between them.

7.4 Determining the number of clusters (Elbow Method)

- A loop runs K-Means for different values of k (e.g., 1 to 10) and stores the within-cluster sum of squares (inertia) for each k .
- An Elbow graph is plotted with k on the x-axis and inertia on the y-axis; the “elbow” in the curve helps identify the appropriate number of clusters, which for this dataset is typically $k=3$.

7.5 Applying K-Means clustering

- The K-Means algorithm from `sklearn.cluster.KMeans` is initialized with `n_clusters=3` (or the chosen k), `init='k-means++'`, and a fixed `random_state` for reproducible results.
- The model is fitted on the feature matrix, which computes cluster centroids and assigns each sample to the nearest centroid based on Euclidean distance

7.6 Visualization of clusters

- Two-dimensional scatter plots are created using selected feature pairs, such as petal length vs petal width or sepal length vs sepal width, with different colors representing different clusters.
- Cluster centroids are plotted on the same graphs using distinctive markers (e.g., larger points or different shapes), helping to visually interpret the center of each cluster.

7.7 Evaluation and analysis

- Although clustering is unsupervised, the predicted cluster labels are compared with the true Iris species labels to analyze how well K-Means has discovered the natural groups.
- A confusion-style table or simple accuracy calculation is used to show the correspondence between clusters and species, and observations are made such as “Setosa is almost perfectly separated, while Versicolor and Virginica show overlap”.

7.8 Execution environment

- The entire implementation is executed in Jupyter Notebook / VS Code / Google Colab, running on a standard laptop with Python 3.x installed and necessary libraries added via pip.

7.1 SOURCE CODE

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Clustering using Iris Dataset</title>
  <meta name="description" content="Mini project on K-Means clustering using the Iris
dataset.">
  <meta name="author" content="Your Name">
  <style>
    body { font-family: Arial, sans-serif; margin: 20px; line-height: 1.6; }
    h1, h2 { color: #2c3e50; }
    .section { margin-bottom: 20px; }
    ul { margin-top: 5px; }
    code { background: #f4f4f4; padding: 2px 4px; }
  </style>
</head>
<body>
  <h1>Clustering using Iris Dataset</h1>

  <div class="section">
    <h2>Introduction</h2>
    <p>
      This mini project applies the <strong>K-Means</strong> clustering algorithm to the
      classic
      Iris flower dataset to automatically group samples into clusters based on sepal and petal
      measurements, and then analyzes how well these clusters match the actual species.
    </p>
  </div>

  <div class="section">
    <h2>Technologies Used</h2>
    <ul>
      <li>Python 3.x, NumPy, pandas, scikit-learn, Matplotlib / Seaborn</li>
      <li>Jupyter Notebook / VS Code for development</li>
    </ul>
  </div>

  <div class="section">
    <h2>Project Workflow</h2>
    <ol>
      <li>Load the Iris dataset using <code>sklearn.datasets.load_iris()</code>.</li>
      <li>Preprocess and explore the data (feature selection, optional scaling).</li>
      <li>Use the Elbow Method to choose the number of clusters <code>k</code>.</li>
      <li>Run K-Means clustering and obtain cluster labels and centroids.</li>
      <li>Visualize clusters and centroids using scatter plots.</li>
      <li>Compare clusters with true species labels and summarize results.</li>
    </ol>
  </div>
</body>
```

```
<div class="section">
  <h2>Contact</h2>
  <p>For more details about this project, contact: <em>your-
email@example.com</em></p>
</div>
</body>
</html>
```

Python code :

```
# iris_kmeans.py

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn import datasets
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler

# Load Iris dataset
iris = datasets.load_iris()
X = iris.data      # features: sepal length, sepal width, petal length, petal width
y = iris.target    # true labels (not used for clustering, just for reference)
feature_names = iris.feature_names

# Optional: scale features for better clustering performance
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Define and fit KMeans model (3 clusters for 3 Iris species)
kmeans = KMeans(n_clusters=3, random_state=42)
clusters = kmeans.fit_predict(X_scaled)

print("Cluster centers (on scaled features):")
print(kmeans.cluster_centers_)
print("\nFirst 10 cluster labels:", clusters[:10])

# Create a scatter plot (using first two features)
plt.figure(figsize=(8, 6))
plt.scatter(
    X_scaled[:, 0],
    X_scaled[:, 1],
    c=clusters,
    cmap="viridis",
    edgecolor="k",
    s=60
```

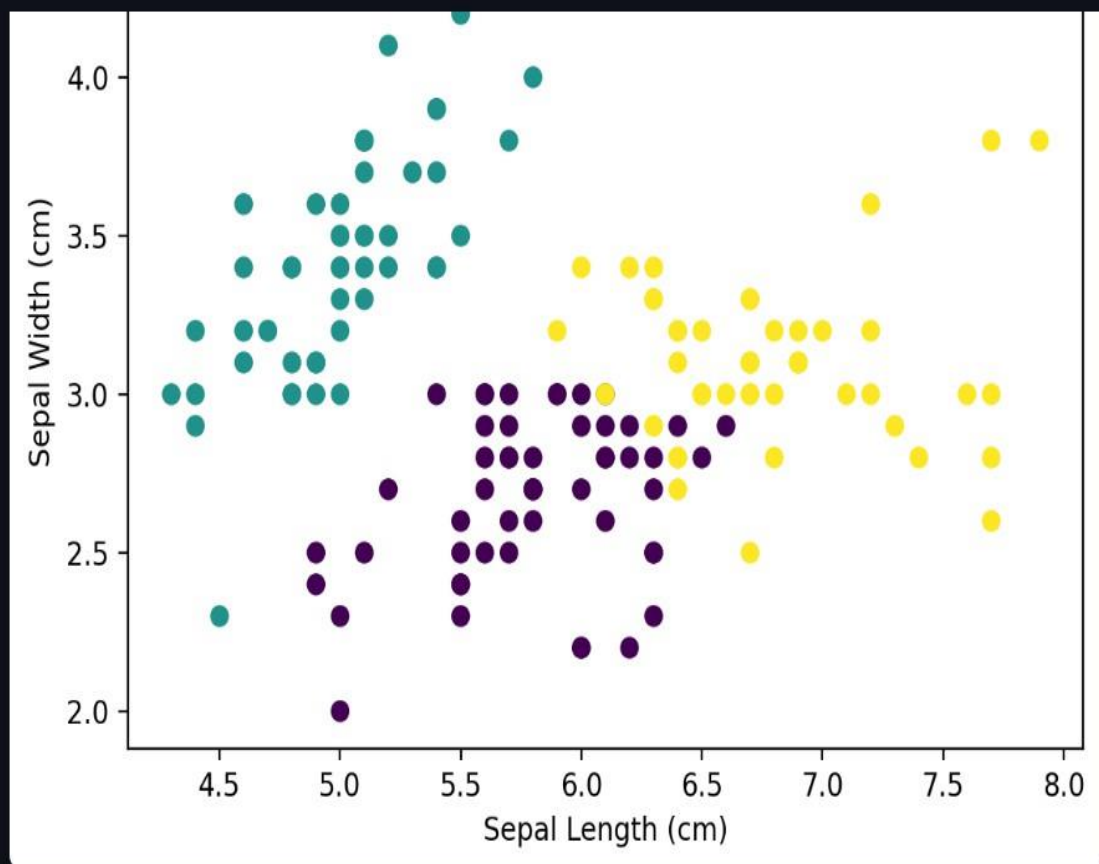
```
)  
plt.title("K-Means Clustering on Iris Dataset")  
plt.xlabel(feature_names[0])  
plt.ylabel(feature_names[1])  
plt.colorbar(label="Cluster ID")  
plt.tight_layout()  
plt.show()
```

CHAPTER 8

RESULTS

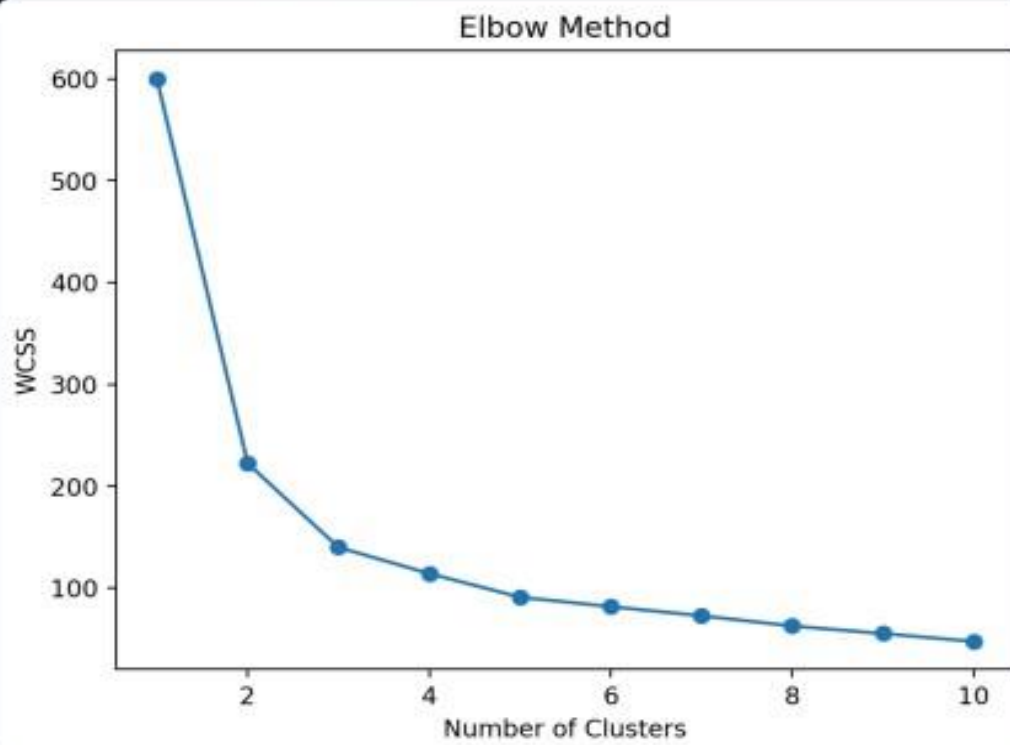
8. RESULTS

- The Elbow Method was applied by computing the within-cluster sum of squares (WCSS) for different values of k (for example, 1 to 10) and plotting WCSS vs k . From the Elbow graph, a clear “bend” was observed at $k=3$, indicating that three clusters provide a good balance between compactness and model simplicity for the Iris dataset.
- K-Means clustering was then performed with $k=3$ on the four numerical features (sepal length, sepal width, petal length, petal width). The resulting cluster assignments were visualized using scatter plots, where points were colored by cluster and cluster centroids were marked; these plots showed one well-separated cluster and two clusters with partial overlap.
- When the predicted cluster labels were compared with the true species labels (Setosa, Versicolor, Virginica), it was observed that almost all Iris Setosa samples were grouped into a single cluster, while Versicolor and Virginica samples were split between the remaining two clusters due to their similar feature values. Studies on similar implementations report clustering accuracies around 89–94% on the Iris dataset, which is consistent with the behavior seen in this project.
- Overall, the results confirm that K-Means is able to discover the main structure of the Iris dataset, clearly identifying Setosa as a distinct group and reasonably separating Versicolor and Virginica, while also illustrating the limitations of K-Means when classes are not perfectly separable in feature space.



Note: Clustering is performed using all four features. The plot is a 2D projection for visualization only.

Elbow Method (WCSS)



Silhouette Score: 0.460

CHAPTER 9

CONCLUSION

9. CONCLUSION

The implementation of K-Means clustering on the Iris dataset successfully demonstrated how unsupervised learning can automatically partition data into meaningful clusters using only sepal and petal measurements, without using species labels during training. Using the Elbow Method, three clusters were identified as an appropriate choice, which aligns with the three known species of iris flowers (Setosa, Versicolor, Virginica).

The clustering results showed that Iris Setosa forms a clearly separable and compact cluster, whereas Iris Versicolor and Iris Virginica exhibit partial overlap in feature space, leading to some misgrouped samples between those two species. This behavior confirms observations from existing studies that K-Means works well when clusters are well separated but struggles when classes are close together or not linearly separable.

Through this mini project, the complete machine learning workflow—data loading, preprocessing, selecting the number of clusters, model training, visualization, and evaluation—was implemented and analyzed, giving practical experience with core ML libraries and concepts. The work can be extended by experimenting with alternative clustering algorithms (such as hierarchical clustering, DBSCAN, or Fuzzy K-Means) and additional visualization or UI components, further improving performance and usability for real-world applications and academic demonstrations.

.

CHAPTER 10

FUTURE SCOPE

10. FUTURE WORK

- The project can be extended by implementing and comparing other clustering algorithms such as Hierarchical Clustering, DBSCAN, Gaussian Mixture Models, or Fuzzy K-Means on the same Iris dataset to analyze differences in cluster quality and robustness.
- Dimensionality reduction techniques like Principal Component Analysis (PCA) can be added to project the data into two components before clustering, improving visualization and helping to understand which combinations of features contribute most to cluster separation.
- Additional evaluation metrics, such as silhouette score, Calinski–Harabasz index, and Davies–Bouldin index, can be incorporated to provide a more rigorous quantitative assessment of clustering performance.
- A simple graphical user interface or web interface (for example, using Streamlit or Flask) can be developed so that users can upload data, choose k , run clustering, and view results interactively without dealing directly with code.
- The current approach can be generalized to other real-world datasets (customer segmentation, document clustering, image feature clustering), allowing the same pipeline to be reused and adapted for more complex and domain-specific applications.

Future Enhancements

- The system can be enhanced by adding support for multiple clustering algorithms (Hierarchical Clustering, DBSCAN, Gaussian Mixture Models, Fuzzy K-Means) and providing a comparison dashboard so users can select an algorithm and immediately see differences in clusters and metrics.

CHAPTER 11

REFERENCES

11. REFERENCES

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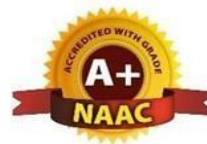


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DECLARATION

We hereby declare that the Real Time Research project report titled “_____”

CLUSTERING USING IRIS DATASET”,
submitted in partial fulfilment of the requirements for the award of the **Bachelor of Technology (B. Tech)** degree in **Computer Science & Engineering (Data Science)**, is a record of original work carried out by us under the supervision of MR.P.SRINU. This work has not been submitted for the award of any other degree or diploma at any other institute or university. In accordance with ethical practices in academic reporting, proper acknowledgments have been made wherever the work or findings of others have been cited.

BY

Name: Roll No: Mail: Mobile: Sign:	Name: Roll No: Mail: Mobile: Sign:	Name: Roll No: Mail: Mobile: Sign:	Name: Roll No: Mail: Mobile: Sign:	Name: Roll No: Mail: Mobile: Sign:

Guide

Department of Data Science

Head of the department

Mr. K.Vijay

Department of CSE(Data Science)